

VEDANT GANDHI

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EXPERIENCE

March Networks - DevOps Software Designing Specialist Co-op | Ottawa, ON

Dec 2024 - Sep 2025

- **Developed** object-oriented **Python** tools using **REST APIs** to analyze and manage release workflows, improving traceability and reducing manual release errors by **30%** across Linux-based software and firmware systems.
- **Engineered** automated validation frameworks to orchestrate regression suites, parse kernel logs, and evaluate **runtime telemetry**, optimizing platform stability and troubleshooting velocity via scripts integrated within **Jenkins CI/CD pipelines**.
- **Designed** reusable libraries utilizing **inheritance and abstraction** for modular data collection, system validation, and diagnostic analysis, significantly bolstering the **reliability** testing and **observability** of complex distributed software systems.
- **Architected** Jenkins-driven pipelines to automate **AWS user telemetry** extraction on the first of every month, streamlining critical data collection workflows and reducing manual reporting overhead by **45%** across all internal departments.

March Networks - QA Test Engineer Co-op | Ottawa, ON

Sep 2024 - Dec 2024

- **Investigated** complex system-level failures across **Linux**-based embedded platforms, identifying **35+** critical defects in C/C++ firmware and camera processing pipelines and **Python test scripts** to verify resolutions and document results.
- **Performed** root-cause analysis using system logs and runtime traces to identify failure conditions, validating engineering fixes by integrating custom scripts with Cypress automation methodologies to increase regression test coverage by **25%**.
- **Recreated** production issues in controlled environments to isolate behaviors, having a **95%** average reproducibility rate for high-priority bugs, accelerating developer defect resolution by **30%** via **streamlined diagnostics** and root-cause analysis.
- **Orchestrated** cross-functional testing strategies to validate hardware-software integration, leveraging scripts to audit system health and enhancing final product reliability by **20%** across **high-priority embedded camera firmware releases**.

CDAC- Front-End Web Applications & Interface Development | India

Jan 2023 - May 2023

- **Designed** and developed responsive web interfaces using **HTML, CSS, and JavaScript** to build user-friendly dashboards and web applications for enterprise and internal digital platforms supporting modern browser compatibility.
- **Implemented** dynamic UI components and integrated REST APIs using JavaScript and React, improving application **usability, navigation flow, and overall front-end performance** across multiple devices and screen sizes.
- **Collaborated** with backend developers and UI/UX designers in Agile workflows, used Git for version control, performed debugging and testing and ensured reliable deployment of **stable and maintainable web applications**.

PROJECTS

Smart Mirror AI - Intelligent Edge & AI System (C, C++, Python, Linux, AI/ML, Automation)

- **Architected** a modular ecosystem integrating C/C++ kernels with **Python** microservices to deliver AI-driven fashion analytics and comparative recommendations via an automated chatbot interface optimized for edge processing.
- **Implemented** Linux modules to orchestrate high-frequency data pipelines and manage strict timing constraints, optimizing resource utilization for **real-time inter-component** communication and deterministic processing.
- **Developed** Python-based **analytical workflows** integrating machine learning **heuristics** and **adaptive decision logic** to dynamically modulate system behavior based on real-time environmental telemetry and performance profiling.
- **Built** automated **test validation suites** to audit system outputs and profile performance metrics, accelerating the resolution of complex integration bottlenecks across distributed low-level software components.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, Bash, Linux

Backend & Full Stack: Fast API, Flask, Node.js, React.js, HTML/CSS, SQL, PostgreSQL, MySQL, MongoDB

Cloud & Infra: Azure, Docker, Kubernetes, Ansible, Datadog, Terraform, Crossplane, GCP, GitHub Actions, AWS

Test & DevOps: Visual Studio, Jenkins, Bitbucket, Jira & Confluence, Postman, CI/CD, Git, GitHub, Scrum, Agile, Power BI

EDUCATION

Carleton University - BCS (Honours), Co-op Option | Ottawa, ON | 3.9 GPA

Sep 2022 – Apr 2027

- Concentration in Artificial Intelligence and Machine Learning with **\$21,000** scholarship for maintaining above **95% average**.
- **Relevant Coursework:** Algorithm Design & Analysis, Operating Systems, Database Management, Web Development

AVAILABILITY : Full-time starting May 1st